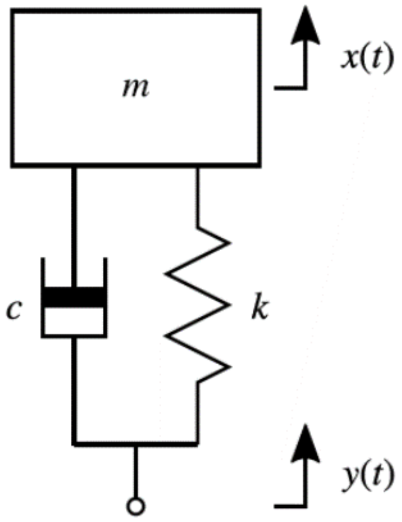
Exercise: Poles, Zeros, Characteristic Equation



$$G(s) = \frac{X(s)}{Y(s)} = \frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ζ ✓

ω_n ✓

Find poles and zeros:

i) Case a: overdamped ($\zeta = 1.25$)

ii) Case b: underdamped ($\zeta = 0.4$)

Note, $\omega_n = 4$ rad/s

a) $$G(s) = \frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

a) $\zeta = 1.25$; $\omega_n = 4$

$$G(s) = \frac{2 \times 1.25 \times 4\, s + 4^2}{s^2 + 2 \times 1.25 \times 4 \times s + 4^2} = \frac{10s + 16}{s^2 + 10s + 16}$$

zeros

characteristic eq^n

poles

$10s + 16 = 0 \Longrightarrow$ zeros $\Rightarrow s = -16/10 = -1.6$

$s^2 + 10s + 16 = 0 \Longrightarrow$ poles $\Rightarrow s = -8$ & $s = -2$

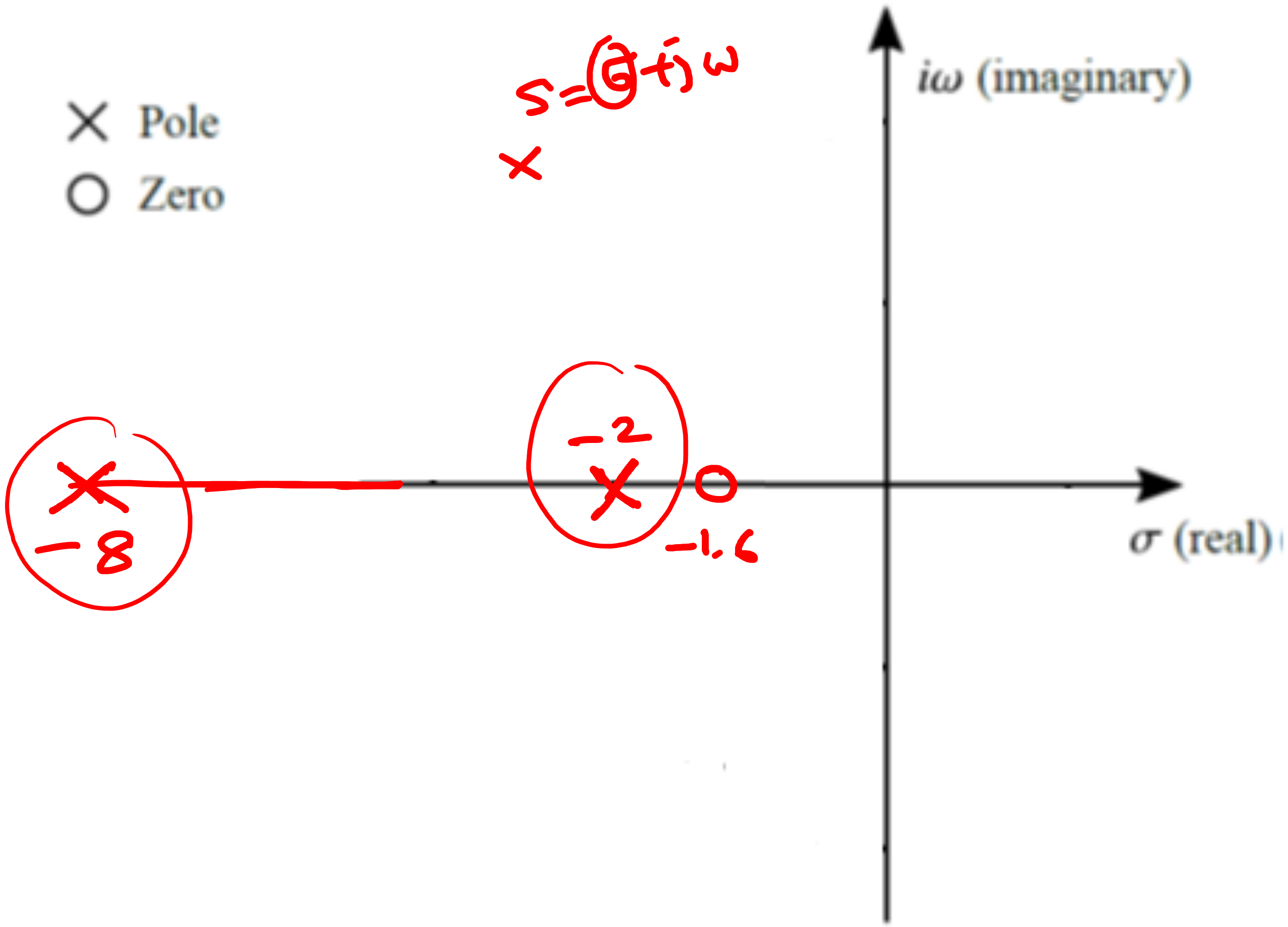
$(s+8)(s+2) = 0$

b) $\zeta = 0.4$; $\omega_n = 4$ rad/s

$$G(s) = \frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{2 \times 0.4 \times 4 + 4^2}{s^2 + 2 \times 0.4 \times 4 + 4^2} =$$

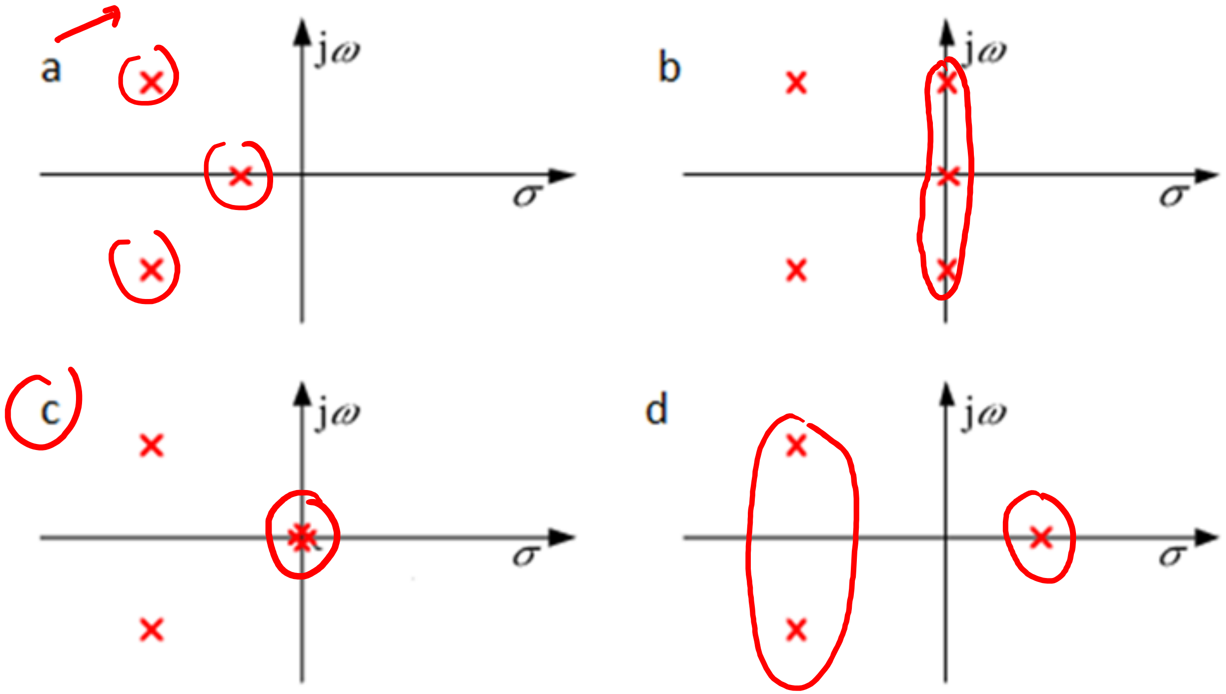
Complex Plane (s-plane)



$\xi = 1.25$
$\Downarrow$
stable.

Exercise

Consider the below s plane pole plots and comment on the expected form of stability for each system.



a) all poles are on LH s-plane ⇒ stable.

b) marginally stable

c) " "

d) unstable.

Exercise: Routh's Stability Criterion

Consider the below polynomial:

$$a(s) = s^6 + 4s^5 + 3s^4 + 2s^3 + s^2 + 4s + 4$$

Determine the stability of the system.

$$a(s) = s^6 + 4s^5 + 3s^4 + 2s^3 + s^2 + 4s + 4$$

⇒ we have all the coefficient   all the coefficient (+) ve ✓

s^6	:	1	3	1	4
s^5	:	4	2	4	0
s^4	:	2.5	0	4	
s^3	:	2	$-12/5$	0	
s^2	:	3	4	0	
s^1	:	$-76/15$	0		
s^0	:	4			

$$-\frac{\begin{vmatrix}1 & 3\\ 4 & 2\end{vmatrix}}{4} = 2.5$$

$$-\frac{\begin{vmatrix}1 & 1\\ 4 & 4\end{vmatrix}}{4} = \frac{-0}{4}$$

$$-\frac{\begin{vmatrix}1 & 4\\ 4 & 0\end{vmatrix}}{4} = \frac{16}{4}$$

$$-\frac{\begin{vmatrix}1 & 0\\ 4 & 0\end{vmatrix}}{4} = 0$$

$$-\frac{\begin{vmatrix}2.5 & 0\\ 2 & -12/5\end{vmatrix}}{2}$$

$$-\frac{\begin{vmatrix}4 & 2\\ 2.5 & 0\end{vmatrix}}{2.5} = \frac{5}{2.5}$$

$$-\frac{\begin{vmatrix}2.5 & 4\\ 2 & 0\end{vmatrix}}{2}$$

$$-\frac{\begin{vmatrix}4 & 4\\ 2.5 & 4\end{vmatrix}}{2.5} = \frac{-(16-10)}{2.5}$$

$+3 \to -76/15 \to +4$

⇒ unstable.

⇒ 2 poles on RHS